

Bank Analytics - Loan Default Prediction System

Project Overview

This project is an end-to-end Machine Learning application that predicts whether a customer is likely to default on a loan. The project includes data preprocessing, exploratory data analysis (EDA), feature engineering, model building, evaluation, and deployment using Streamlit.

The application allows users to enter customer details through a web interface and instantly predicts the loan default risk.

Objectives

- Analyze banking customer data.
 - Identify factors influencing loan default.
 - Build a machine learning model for loan risk prediction.
 - Deploy the model using Streamlit.
 - Provide real-time predictions through an interactive web application.
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Project Structure

Bank_Analytics_App/

— Bank_App.py	# Streamlit Application
— Bank_analytics_model.pkl	# Trained Machine Learning Model
— requirements.txt	# Required Python Packages
— README.md	# Project Documentation

Technologies Used

- Python
- Pandas
- NumPy
- Scikit-learn
- Streamlit

- Joblib
 - Matplotlib
 - Seaborn
 - Jupyter Notebook
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Dataset

The dataset contains banking information such as:

- Customer Information
- Account Details
- Loan Details
- Account Information
- Order Details
- Transaction History
- Card Information
- District Information

The final dataset contains engineered features used for predicting loan default.

Feature Engineering

The following features were used for model building:

Numerical Features

- Age
- Average Balance
- Number of Transactions
- Total Credit
- Total Withdrawal
- Has Card
- Loan Duration
- Monthly Payments

Categorical Features

- Gender
- Statement Frequency

- Account Type
 - Loan Group
 - Loan Purpose
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Machine Learning Model

Model Used:

- Logistic Regression

Pipeline Includes:

- Missing Value Handling
 - StandardScaler
 - OneHotEncoder
 - Logistic Regression Classifier
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Model Performance

Metric	Score
Accuracy	80.91%
Precision	25.27%
Recall	83.51%
F1 Score	38.80%
ROC-AUC	91.42%

The model achieved a high ROC-AUC score, demonstrating good capability in distinguishing between default and non-default loans.

Streamlit Web Application

The web application includes:

- User Login
- Customer Information Form
- Loan Default Prediction
- Default Probability

- Risk Classification
 - User-Friendly Interface
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Installation

Clone the repository

```
git clone <repository-url>
```

Move into the project folder

```
cd Bank_Analytics_App
```

Install dependencies

```
pip install -r requirements.txt
```

Run the application

```
streamlit run Bank_App.py
```

Application Workflow

1. Login to the application.
 2. Enter customer details.
 3. Click **Predict**.
 4. View loan default prediction.
 5. Check default probability and risk level.
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Future Improvements

- XGBoost and Random Forest models
- Explainable AI (SHAP)
- PDF prediction reports
- Database integration
- User authentication with a database
- Cloud deployment
- Dashboard analytics
- REST API integration



Author

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- MCA (Generative AI)
 - BCA (Data Science)
 - Python | SQL | Power BI | Machine Learning | Streamlit
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Acknowledgements

This project was developed for educational purposes to demonstrate the complete machine learning lifecycle, from data preprocessing and model development to deployment as an interactive web application.